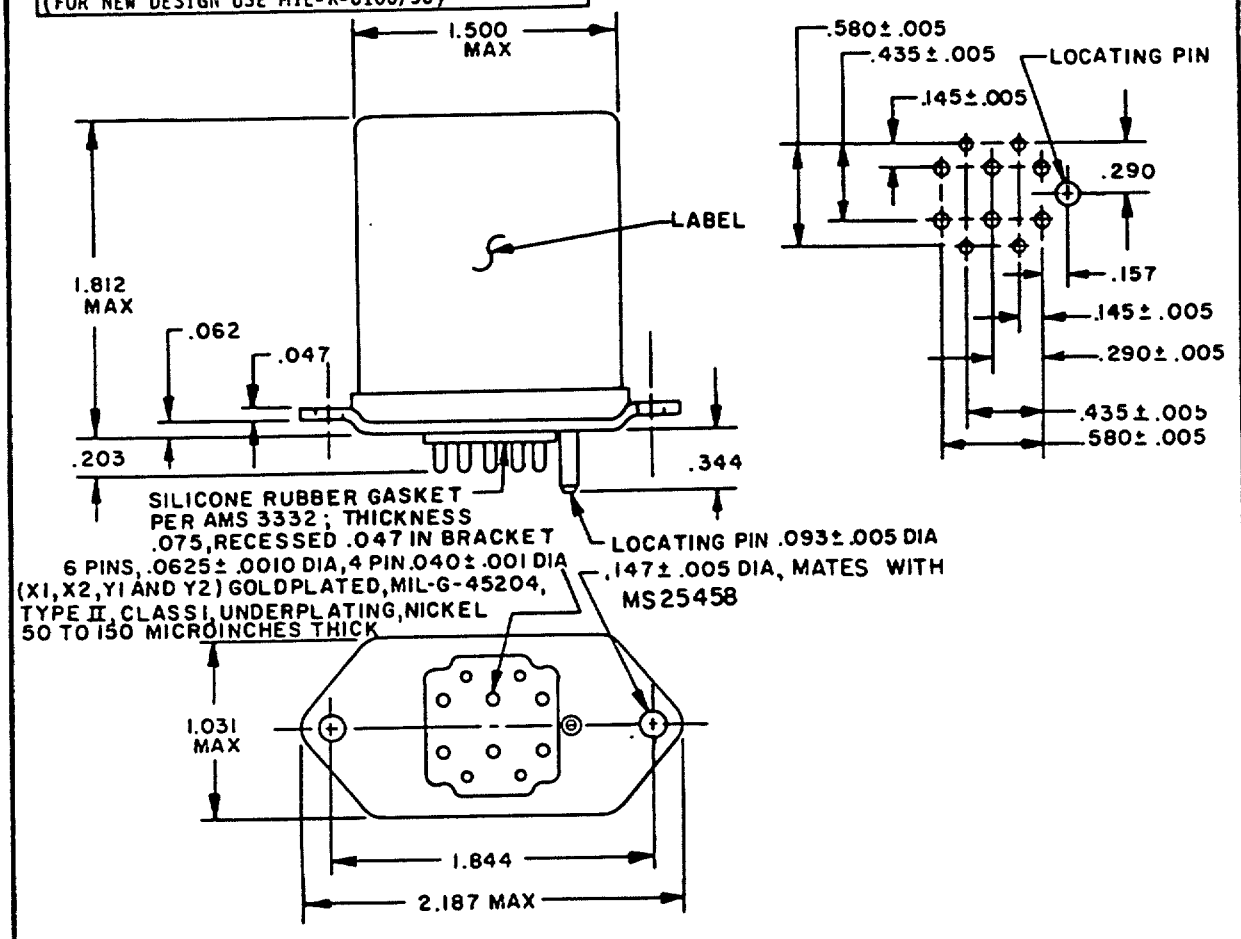
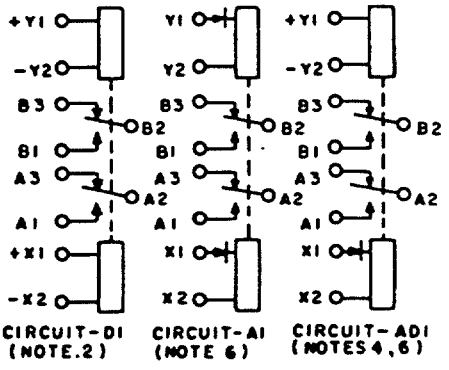
FED. SUP CLASS
5945

INACTIVE FOR NEW DESIGN AFTER 5 JUN 87
NO SUPERSEDING STANDARD.
(FOR NEW DESIGN USE MIL-R-6106/50)



Inches	mm	Inches	mm
.001	0.03	.203	5.16
.005	0.13	.293	7.74
.040	1.02	.344	8.74
.047	1.19	.435	11.05
.062	1.57	.580	14.73
.0625	1.59	1.031	26.19
.093	2.36	1.500	38.10
.145	3.68	1.812	46.02
.147	3.73	1.844	46.84
.157	3.99	2.187	55.55



(F) ENTIRE STANDARD REVISED

APPROVED 1 Nov 1960 REVISED (F) 5 JUN 87

User activities: Army - Navy - Air Force -
Review activities: Army - EC Navy - Air Force - 11

This military standard is approved for use by all Departments and Agencies of the Department of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

P.A AF - 85 Other Cust Navy - AS	International interest	TITLE RELAYS, ELECTROMAGNETIC, 10 AMPERES, 2 PDT, TYPE I, MAGNETIC LATCH, SOCKET MOUNTED, HERMETICALLY SEALED	MILITARY STANDARD MS25457
Procurement Specification MIL-R-6106		SUPERSEDES:	PAGE 1 OF 5

DD FORM 1 MAY 73 672 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE
AMSC N/A

5945-0745-16

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm).
4. Relay is magnetically latched in both positions. Caution note to observe polarity must appear on relays with dc coils.
5. Terminal numbers need not appear on relay headers provided there is affixed to the relay a suitable legible circuit diagram that permanently and positively identifies each terminal location specified herein.
6. The use of diodes on ac relays is optional. Actual application must be shown on label.
7. Shock, vibration, and acceleration requirements application with coils de-energized.
8. In the event of confilict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
9. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

TABLE I. Dash numbers and characteristics.

Dash number MS 25457-	Type	Coil	Terminal type	Max weight in pounds
D1	I	dc	Plug in	0.30
A1	I	ac	Plug in	0.32
AD1	I	ac-dc	Plug in	0.32

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Review activities: Army - EC Navy - Air Force - 11
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Review activities: Army -
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Air Force -

TABLE II. Operating characteristics.

MS part no. MS25457-	Coil data												Time (milliseconds max)			
	Coil	Nominal			Max		Max pick-up voltage			Drop-out voltage	Operate 4/	Release 5/	Contact bounce			
													Main		Aux	
		Volts 1/	Freq. Hz	Res Ω	Volts	Amperes	Nominal 3/	High temp test	Cont current test				NO	NC	NO	NC
D1	X1, X2 Y1, Y2	28	dc	N/A	29	0.17	18	18	19.8	N/A	25	N/A	2	2		
A1	X1, X2 Y1, Y2	115	400 2/	N/A	122	0.07	90	90	95	N/A	25	N/A	2	2		
AD1	X1, X2	115	400 2/	N/A	122	0.07	90	90	95	N/A	25	N/A	2	2		
	Y1, Y2	28	dc	N/A	29	0.17	18	18	19.8	N/A	25	N/A	2	2		

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ MS25457-A1 and AD1 may be used on 60 Hz if maximum ambient temperature is limited to +85°C (maximum coil current shall be 0.077 ampere).
3/ Over the temperature range.
4/ With nominal coil voltage.
5/ From nominal coil voltage.

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TABLE III. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles X 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	100	10	10			10	6			10	6			
Inductive	100													
Inductive	20	6	6			10	4			10	4			
Motor	100	4	4			4	3			4	3			
Lamp	100	2	2			2	1.5			2	1.5			
Transfer load														2/
Mechanical life reduced current	400	2.5	2.5			2.5	2			2.5	2			
Intmd current	Applicable per specification													

1/ Absence of value indicates relay is not rated for 3-phase applications.
2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

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Review activities: Army -
Navy - EC
Air Force - 11

User activities: Army -
Navy -
Air Force -

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P.A AIR FORCE 85
Other Cust NAVY - AS

International interest

Procurement Specification MIL-R-6106

SUPERSEDES:

TITLE RELAYS, ELECTROMAGNETIC, 10 AMPERES, 2 PDT, TYPE 1, MAGNETIC LATCH, SOCKET MOUNTED, HERMETICALLY SEALED

MILITARY STANDARD

MS25457

PAGE 5 OF 5

FED. SUP CLASS 5945

Environmental characteristics

Temperature range	-70°C to +125°C
Max altitude rating	80,000 ft
Shock G-level	50 G
Duration	11 ms
Max duration contact opening	10 µs
Vibration - sinusoidal	
G-level	10 G
Frequency range	5 - 1500 Hz
Acceleration	15 G

Electrical characteristics

Insulation resistance, initial	100 megohms
After life or environmental tests	50 megohms

Dielectric strength initial (sea level)

	Initial	After life tests
Coil to case	1,000 V rms	1,000 V rms
Aux contacts		
All other points	1,500 V rms	1,125 V rms

Dielectric strength (altitude)	(When mounted in mating socket) 80,000 ft
Coil to case	500 V rms
Aux contacts	
All other points	500 V rms

Max contact drop initial	0.150 volt
After life test	0.175 volt
Overload current	40 amperes dc, 60 amperes ac
Rupture current	50 amperes dc, 80 amperes ac
Duty rating	Continuous
RFI specification	MIL-STD-461

(Applicable to coil circuits of ac operated relays)

Quality conformance inspection

Performance of groups B and C tests are not applicable.

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